

Why four NeuronBench symbolic matches are false positives

Follow-up to runs 313195 and 313196

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Answer. The four Pareto rows with $10^{-3} < \text{NRMSE} < 0.05$ are not genuine recoveries of the physical ground-truth equation. All four omit the variable input I_{ext} ; three also omit the standalone leak term $-0.3V$. They pass only because the current SRBench-style checker rounds their target-scaled coefficients and then accepts a constant symbolic difference.

The four rows

Every one has different polynomial support from ground truth.

Run	World	Seed	Test NRMSE	Missing terms	Rounded error
313196	d_type	10000	3.2881×10^{-3}	I_{ext}, V	0.001
313196	d_type	10002	3.2881×10^{-3}	I_{ext}, V	0.001
313196	d_type	10004	3.1933×10^{-3}	I_{ext}	0.002
313195	ca_rebound	10002	3.2264×10^{-3}	I_{ext}, V	0.002

The “rounded error” column is the symbolic difference returned by the project checker. Each row is accepted by its *difference-is-constant* rule; none has zero unrounded difference and none has the full ground-truth support.

Concrete example: d_type, seed 10002

The ground-truth physical equation is

$$f_{\text{GT}} = I_{\text{ext}} - 120V\phi_{\text{Na}} - 36V\phi_{\text{K}} - 9V\phi_{\text{D}} - 0.3V \\ + 6000\phi_{\text{Na}} - 2772\phi_{\text{K}} - 693\phi_{\text{D}} - 16.32.$$

The complexity-27 Pareto equation that the checker labels a match has $\text{NRMSE} = 0.003288$. After undoing target scaling and expanding it, the equation is approximately

$$\hat{f} = -120.196V\phi_{\text{Na}} - 36.163V\phi_{\text{K}} - 9.168V\phi_{\text{D}} \\ + 5997.642\phi_{\text{Na}} - 2775.432\phi_{\text{K}} - 695.790\phi_{\text{D}} - 21.897.$$

Most importantly, $\hat{f}$ has no I_{ext} term and no standalone V term. Its physical difference from the target is approximately

$$\hat{f} - f_{\text{GT}} = -I_{\text{ext}} + 0.3V - 0.196V\phi_{\text{Na}} - 0.163V\phi_{\text{K}} - 0.168V\phi_{\text{D}} \\ - 2.358\phi_{\text{Na}} - 3.432\phi_{\text{K}} - 2.790\phi_{\text{D}} - 5.577.$$

This is plainly not an algebraic equivalent of the governing equation.

Why the checker accepts it

PySR fits the target divided by its training RMS, which is roughly 5,400 for this world. In the normalized equation, the exact coefficients become roughly

$$I_{\text{ext}} : 1.84 \times 10^{-4}, \quad V : -5.53 \times 10^{-5}.$$

The project checker applies its SRBench-inspired floating-point cleanup before symbolic simplification:

1. constants smaller than 10^{-4} are replaced by zero;
2. other floating constants are rounded to three decimal places; and
3. a constant difference is accepted as a symbolic match.

Consequently, the normalized V coefficient is explicitly zeroed and the I_{ext} coefficient rounds to 0.000. Other coefficient discrepancies also vanish at this resolution. What remains is the constant 0.001, and the constant-difference rule declares a match.

Conclusion

These four rows should not count as NeuronBench ground-truth recoveries. The issue is not merely that fitted constants differ in their final decimals: the expressions omit variable-dependent terms from the physical law. The held-out NRMSE correctly exposes the omission, whereas the combination of target scaling, coarse coefficient rounding, and the constant-difference rule produces a symbolic false positive.

For NeuronBench, symbolic checks should operate on canonical expanded equations in physical units, or use scale-aware coefficient tolerances that never erase a ground-truth variable term. Numerical recovery and polynomial-support equality should remain separately reported.

Source rows: the saved Pareto frontiers in `runs/313195/neuron_full_eval/neuron_results.json` and `runs/313196/neuron_full_eval/neuron_results.json`. The checker is `evaluation.check_pysr_symbolic_match`.